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Introduction

A **LAN (Local Area Network)** is a type of network used to connect a group of devices within a limited geographical area such as a room, house, or office. the main purpose of this network is to enable devices to share data and resources such as printers, files, and internet access efficiently and quickly.
In this setup we've created a LAN consisting of a **router** and **three computers** the router is the central device responsible for assigning IP addresses to each device managing data traffic within the network and potentially providing internet access if connected to an external modem.

Network Components:

1. **Router:**
    - Manages the local network and connects it to the internet if needed.
    - Distributes IP addresses to connected devices.
    - Supports both wired (Ethernet) and wireless (Wi-Fi) connections.
2. **Three Computers (PCs):**
    - Connected to the router to form an internal network.
    - Capable of sharing files and printers over the LAN.
    - Can connect via Ethernet for higher speed or via Wi-Fi for flexibility.
3. **Connection Medium (Wi-Fi or Ethernet cables):**
    - Used to transmit data between the devices and the router.
    - Wired connections offer higher stability and speed, while wireless offers mobility.

Advantages of the LAN Network in This Setup:

1. **High Data Transfer Speed:**
   Transferring files between devices is very fast compared to wide area networks (WANs).

2. **Resource Sharing:**
   Any device can share a printer or files with the others without needing physical transfer methods like USB drives.

Disadvantages of the LAN Network:

1. **Limited Range:**
   The network only covers a small physical area. It can't be used over long distances without additional hardware like signal extenders.

2. **Router Dependency:**
   If the router fails, the entire network goes down and the devices can't communicate.

Physical Topology: Star Topology

In this LAN setup the physical topology used is the **Star Topology** In a star topology all devices (in this case the three computers) are individually connected to a central device which is the **router**.

Each computer communicates through the router rather than directly with other computers This central point manages and controls the data traffic in the network.

Why Star Topology?

The star topology is commonly used in modern networks because it is **easy to set up**, **troubleshoot**, and **expand**.

It suits both wired and wireless connections, which makes it ideal for small LANs using routers.

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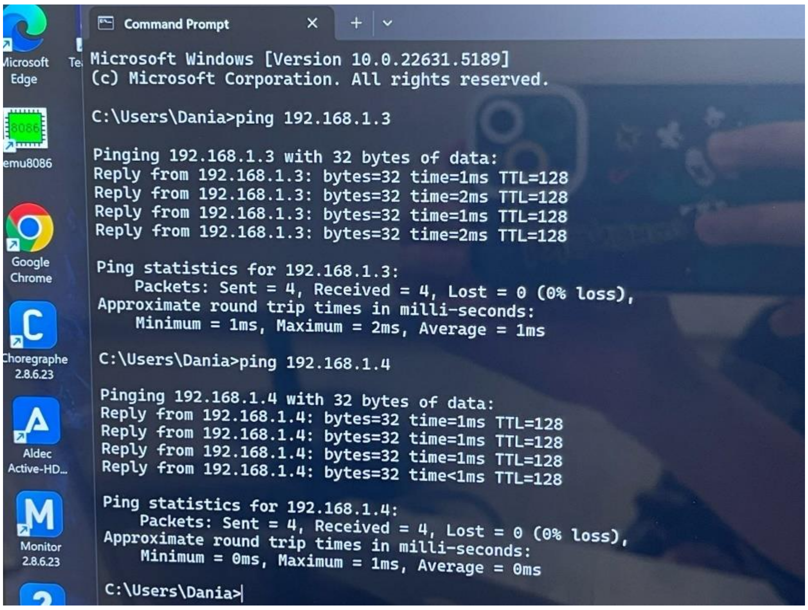
1. Tasks

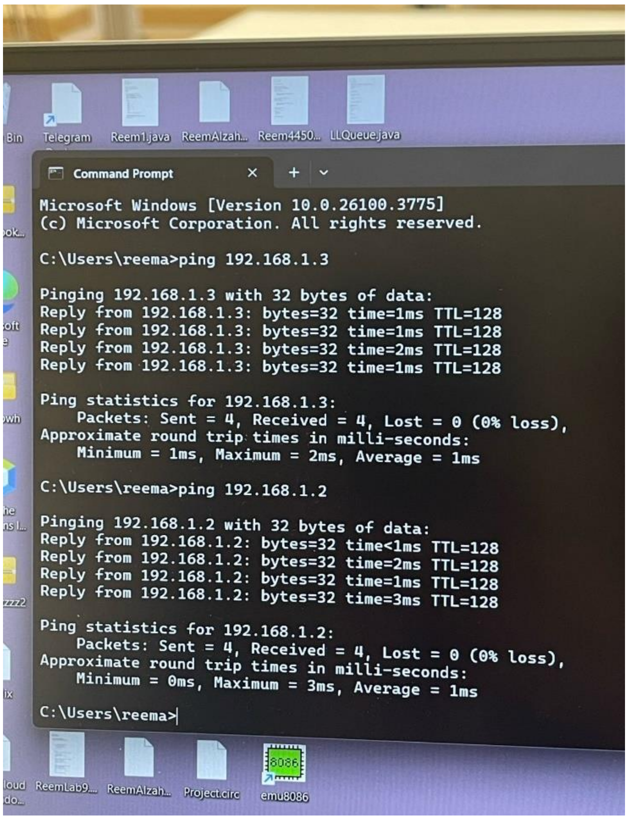
1.1. Performing a ping

Ping is a tool that helps verify whether your device can connect to other devices either on the local network or the internet it works by sending small packets of data and measuring how long it takes to get a response it's often used to diagnose connection issues and check if a specific device is reachable You can access it through the Command Prompt on Windows or the Terminal on other operating systems.

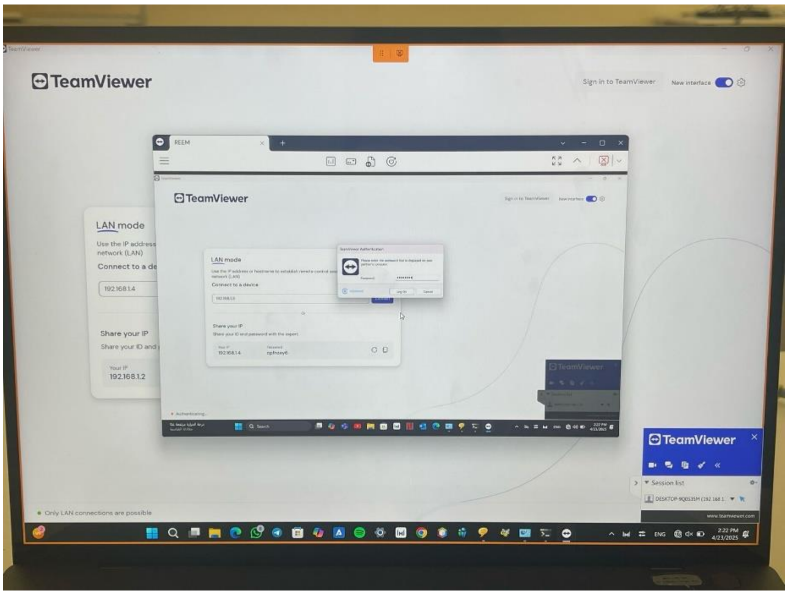
How to Use Ping:

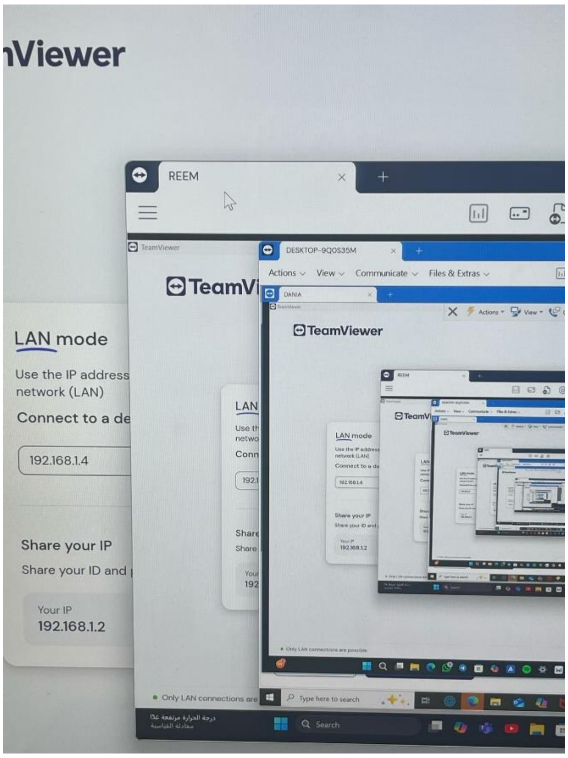
- On Windows: press Windows + R, type cmd and press Enter to open the Command Prompt.
- Then type the ping command followed by the IP address or website you want to test and press Enter.
- You'll see results showing the connection status and the round-trip time of the data packets.


Microsoft Edge
emu8086
Google Chrome
Choregraphe 2.8.6.23
Aldec Active-HD...
Monitor 2.8.6.23
Command Prompt
Microsoft Windows [Version 10.0.22631.5189]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Dania>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\Dania>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\Dania>


Telegram
Reem1java
ReemAlzah...
Reem4450...
LLQueue.java
Command Prompt
Microsoft Windows [Version 10.0.26100.3775]
(c) Microsoft Corporation. All rights reserved.
C:\Users\reema>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\reema>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 3ms, Average = 1ms
C:\Users\reema>
ReemLab9...
ReemAlzah...
Project.circ
emu8086

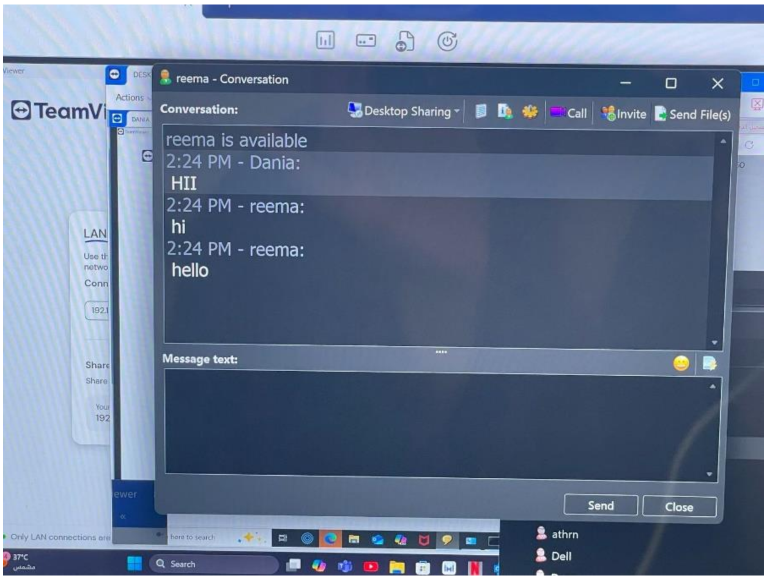
1.2. Control other node using Remote Desktop application

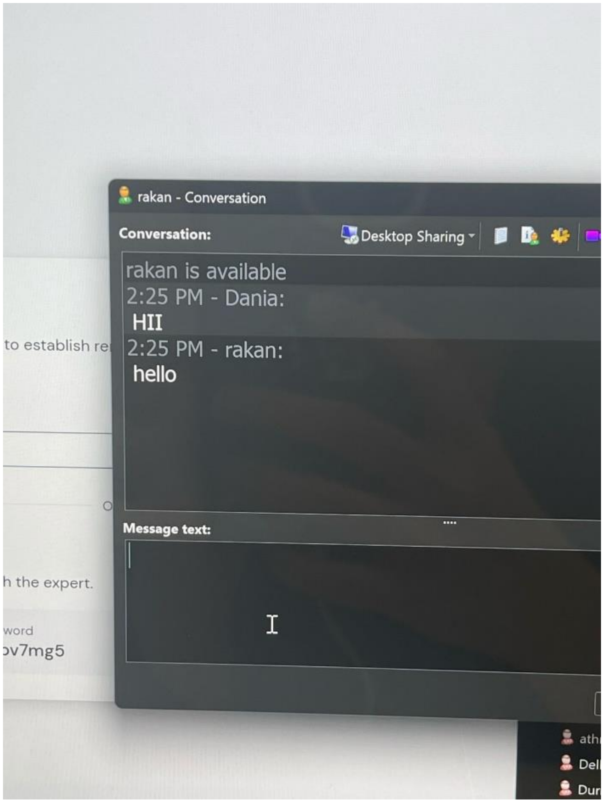




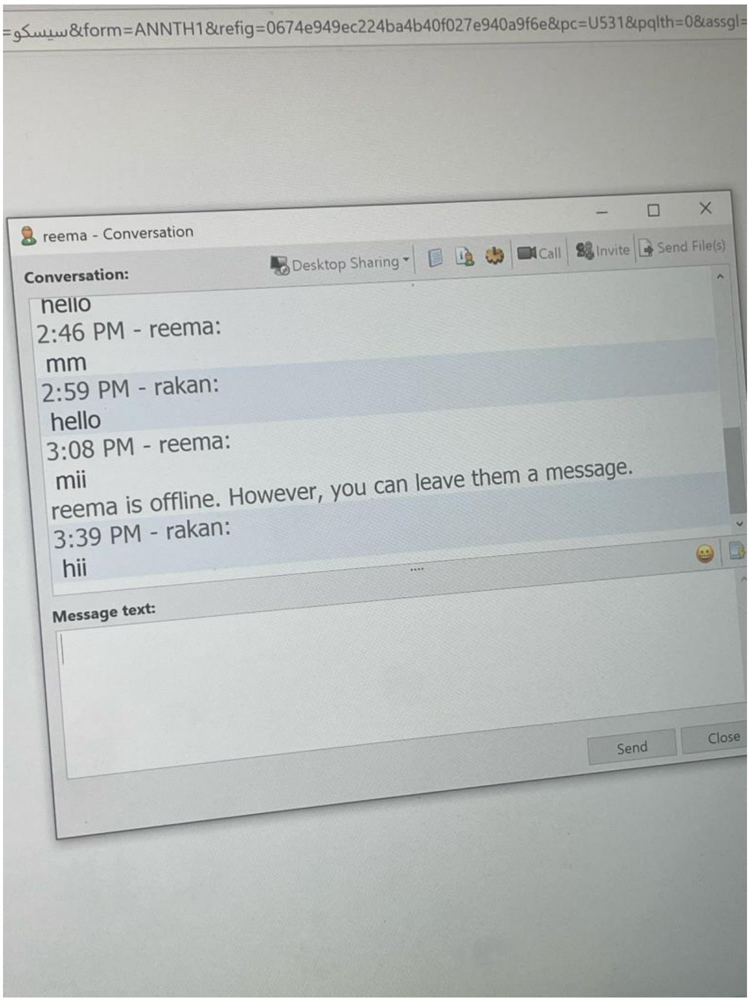
1.3. Basic Firewall Configuration

chat program:





after application is blocked:



Explain how firewalls enhance network security:

Traffic Filtering:
Firewalls inspect all data packets and decide whether to allow or block them based on security rules (IP address, port, or protocol).

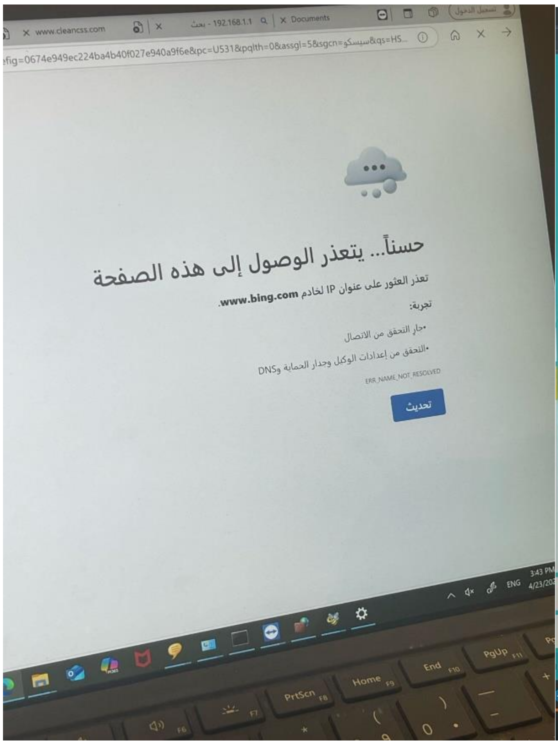
Unauthorized Prevent Access:
Firewalls block hackers or malicious software from accessing devices on your private network.

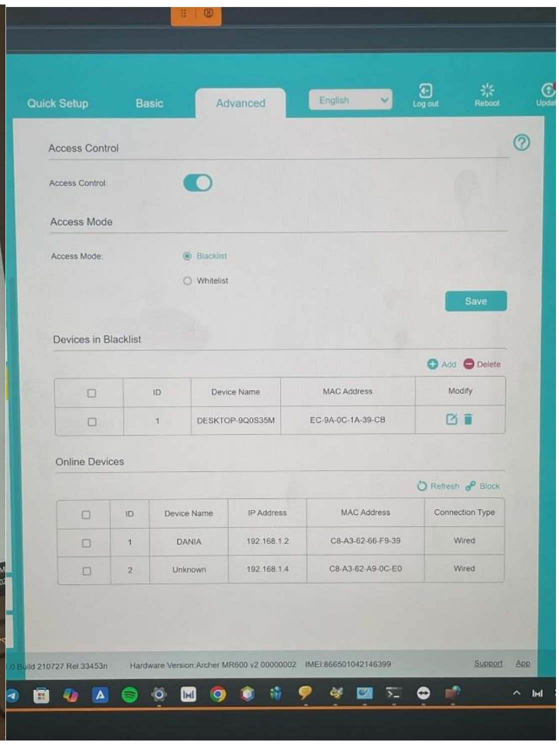
Monitor Network Activity:
They log traffic and can alert administrators to suspicious activity helping detect threats early.

Control Application Access:
Firewalls can restrict which applications or services are allowed to send or receive data preventing unwanted programs from communicating externally.

Prevent Malware Spread:
If one device gets infected a firewall can limit how the malware spreads across the network.

1.4. Implementing Basic Network Security





How Firewall Techniques Prevent Unauthorized Access:

1. Packet Filtering:

- Examines each data packet's header (source IP, destination IP, port, protocol).
- If the packet doesn't match allowed rules it's dropped.

2. Stateful Inspection (Dynamic Filtering):

- Tracks active connections and allows only responses to valid established sessions.

3. Application Layer Filtering:

- Filters traffic based on specific applications or services (HTTP, FTP).
- Can block or allow software like browsers, messengers, or games.